

Free & Open Source Software

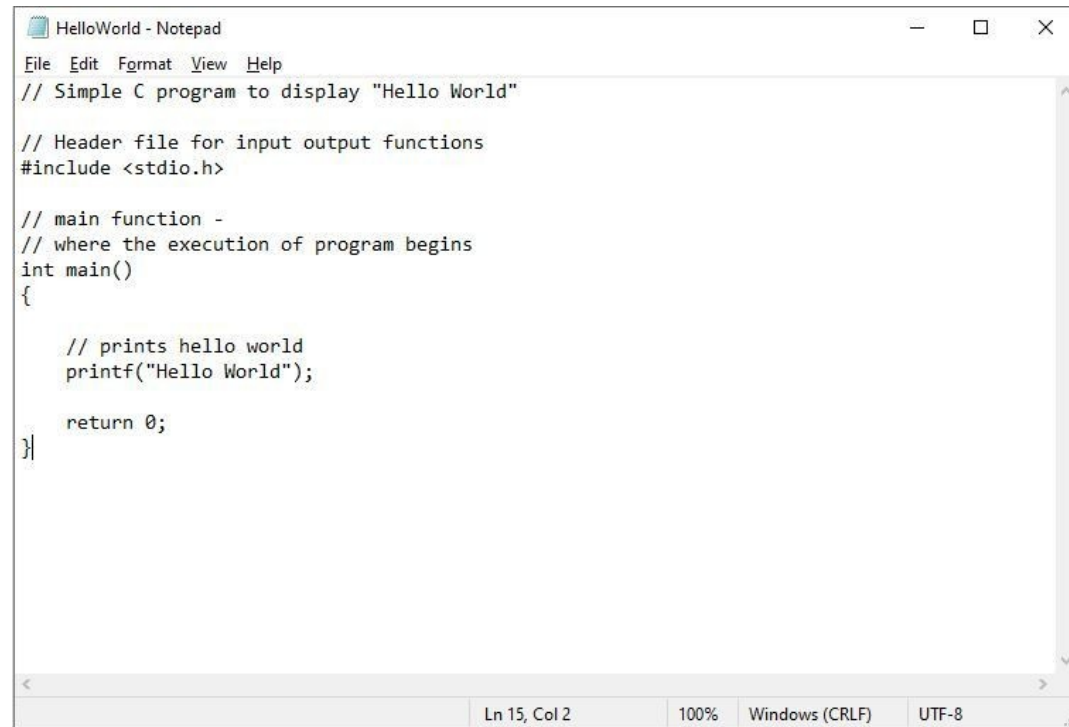
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How are application software built?

The source code

- Human-readable
- A language of some sort



```
File Edit Format View Help
// Simple C program to display "Hello World"

// Header file for input output functions
#include <stdio.h>

// main function -
// where the execution of program begins
int main()
{
    // prints hello world
    printf("Hello World");

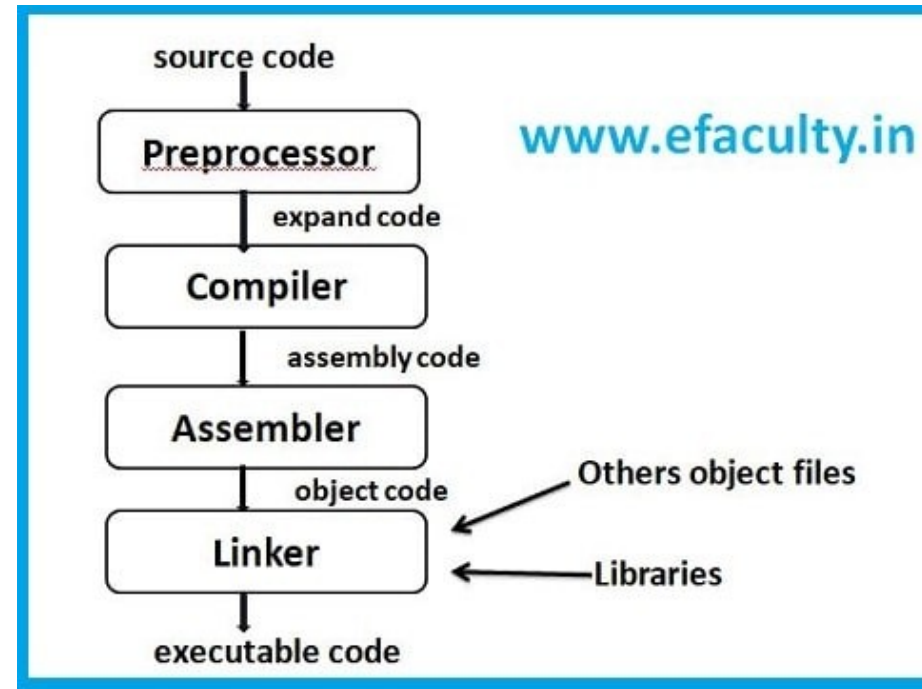
    return 0;
}
```

Ln 15, Col 2 100% Windows (CRLF) UTF-8

How are application software built?

The building process (*sensu lato* compilation)

- Translate the source code to executable
- One way deal, i.e. irreversible process – the exact source code cannot be recreated!



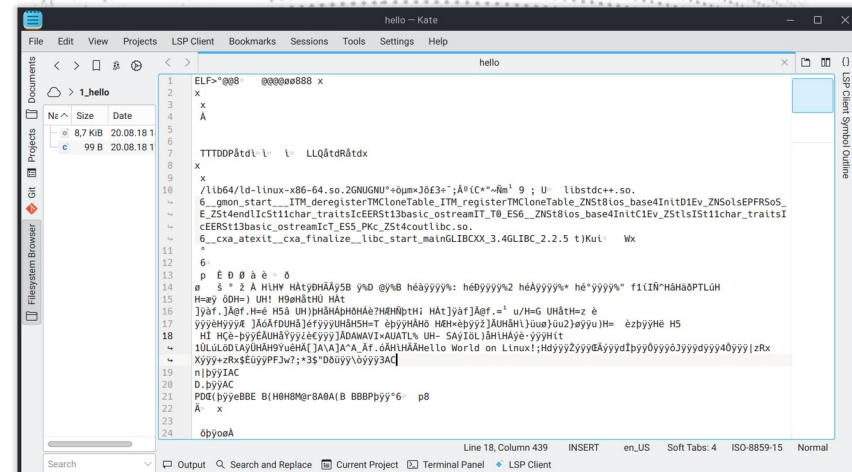
How are application software built?

Result: **binary executable**

- Modification is limited
- What the program does is cryptic (almost black box)
- Specific to Operating System and Architecture!

Download selection

- ☐ gplates_2.3.0_win64.exe
- ☐ gplates_2.3.0_win64.zip
- ☐ gplates_2.3.0_Darwin-x86_64.dmg
- ☐ gplates_2.3.0_ubuntu-18.04-amd64.deb
- ☐ gplates_2.3.0_ubuntu-20.04-amd64.deb
- ☐ gplates_2.3.0_ubuntu-20.10-amd64.deb
- ☐ gplates_2.3.0_ubuntu-21.04-amd64.deb
- ☐ gplates_2.3.0_ubuntu-21.10-amd64.deb
- ☐ gplates_2.3.0_ubuntu-22.04-amd64.deb
- ☐ gplates_2.3.0_ubuntu-22.10-amd64.deb
- ☐ gplates_2.3.0_ubuntu-23.04-amd64.deb
- ☐ gplates_2.3.0_src.zip
- ☐ gplates_2.3.0_src.tar.bz2



The screenshot shows a code editor window titled 'hello - Kate'. The editor displays a binary file named 'hello' with a file size of 99 B. The content is a mix of ASCII characters and non-printable bytes, representing a compiled program. The editor interface includes a menu bar (File, Edit, View, Projects, LSP Client, Bookmarks, Sessions, Tools, Settings, Help), a toolbar, and a sidebar with a file browser and project explorer. The status bar at the bottom indicates 'Line 18, Column 439', 'INSERT' mode, 'en_US' locale, 'Soft Tabs: 4', 'ISO-8859-15' encoding, and 'Normal' line endings.

Free and Open Source Software?

Result

- You don't need to use binaries from the authors (no charge or restrictions)
- You can modify the program's behavior
- You can see what the program does



SOURCEFORGE



Original paradigm

Software only for specific hardware!

- No transferability
- Apple still does this



Same Hardware → Different Software

- Proprietary operating systems
- Expensive, opaque
- **UNIX** (1969, AT&T Bell Labs)



Ken Thompson and
Dennis Ritchie

UNIX®

A Standard of The Open Group®

AIX

XENIX

solaris™

BSD

hp ux



It's a UNIX system.
I know this.

A Free operating system?

Richard Stallman

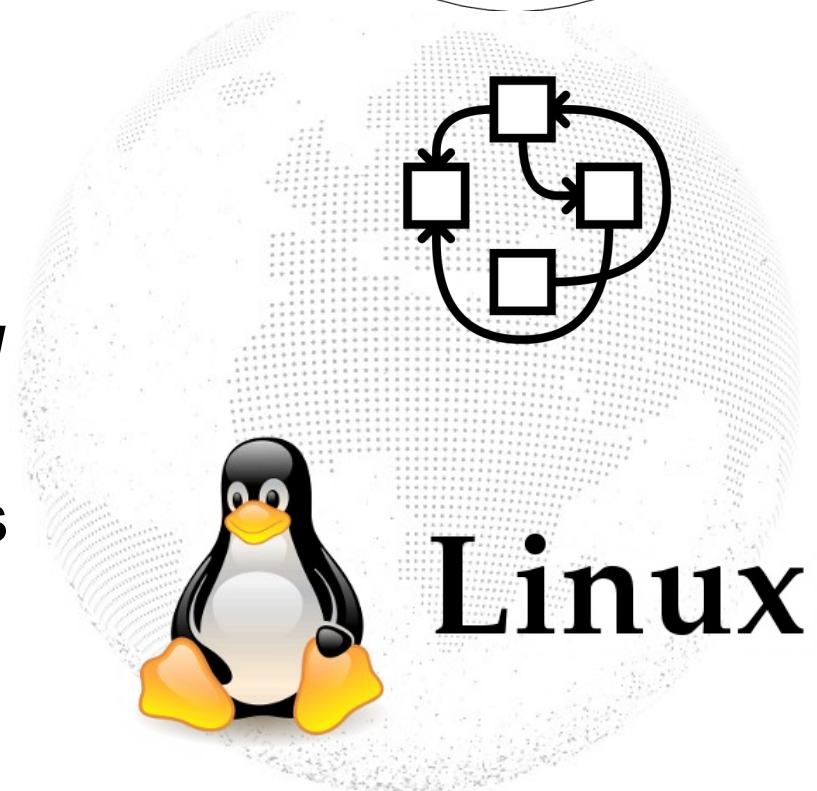
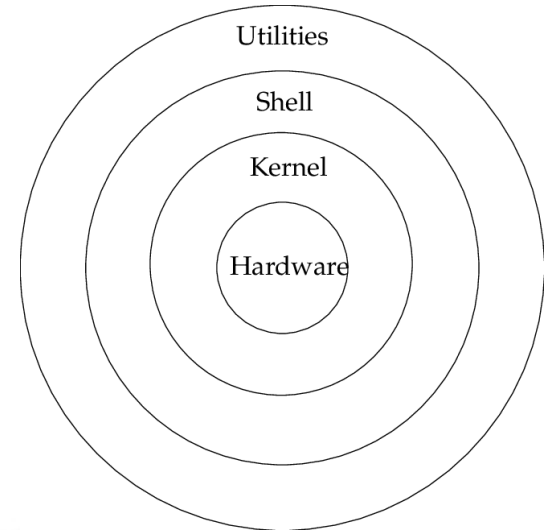


- @MIT: **GNU** is **Not UNIX** (1983)
- Unix-like OS: Modular design
- do one thing, but very good!
- Hundreds of software (including R!!)
- Works well with other open source software

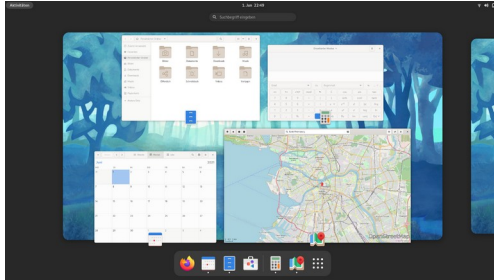


The Kernel

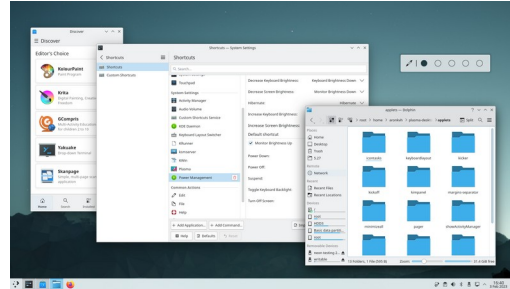
- The most important package of the OS, is built around this:
Windows uses **NT**, MacOS: **Darwin**
- Handles hardware resources
- Original plans for GNU: *Hurd*
- 1991 Linux by Linus Torvalds



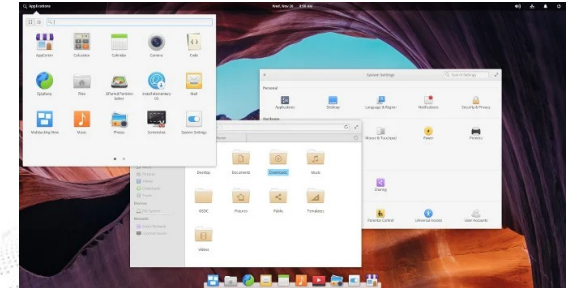
The Desktop Environment



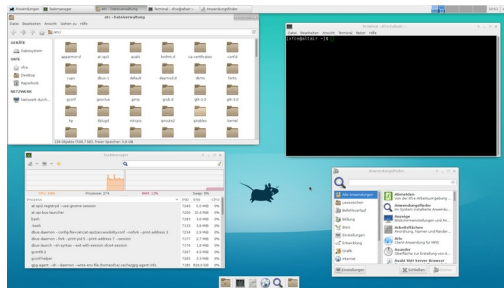
GNOME



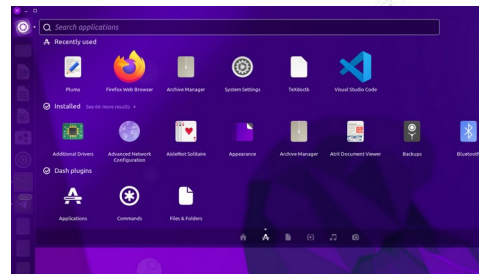
KDE



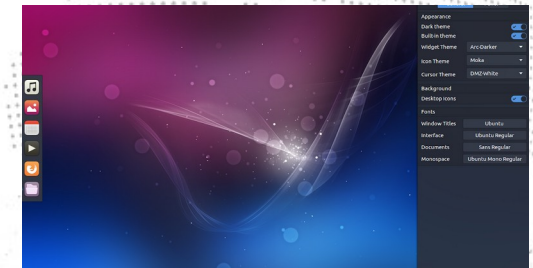
Pantheon



XFCE



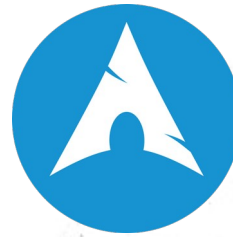
Unity



Budgie

Package management

- You can build programs yourself, but it is easier to use pre-built ones
- Most important/prevalent ones



Primary Distribution	Debian	Arch	Red Hat
Manager Program	dpkg/apt	pacman	Re
Package extension	.deb	(AUR)	.rpm

The Phylogeny

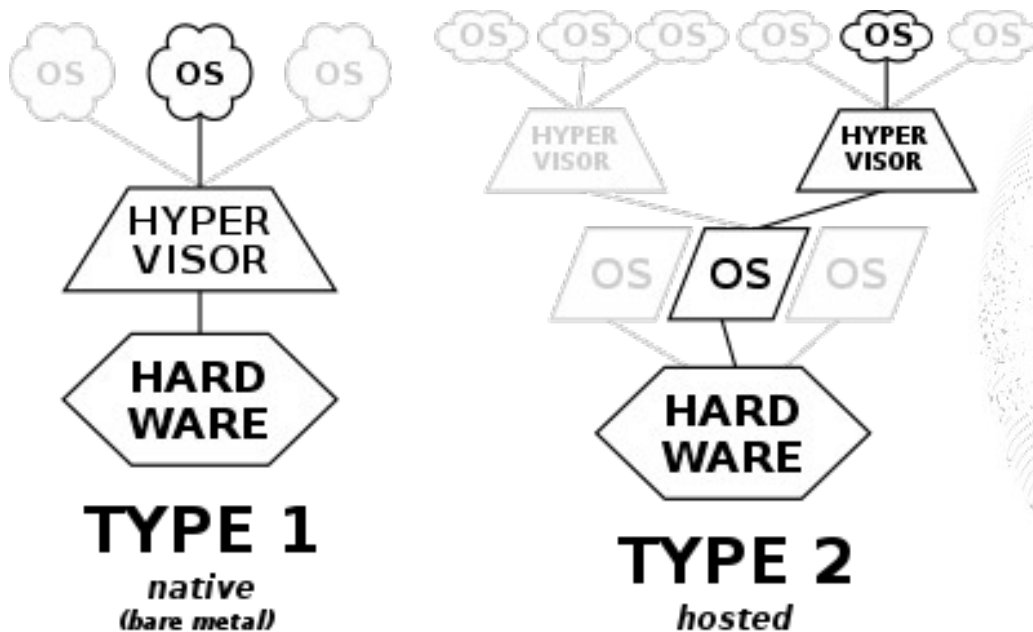
[https://en.wikipedia.org/wiki/Linux_distribution#/media/
File:2023_Linux_Distributions_Timeline.svg](https://en.wikipedia.org/wiki/Linux_distribution#/media/File:2023_Linux_Distributions_Timeline.svg)



Try them!

In virtual computers...

<https://www.youtube.com/watch?v=v1JVqd8M3Yc>



Inkscape

Vector graphics



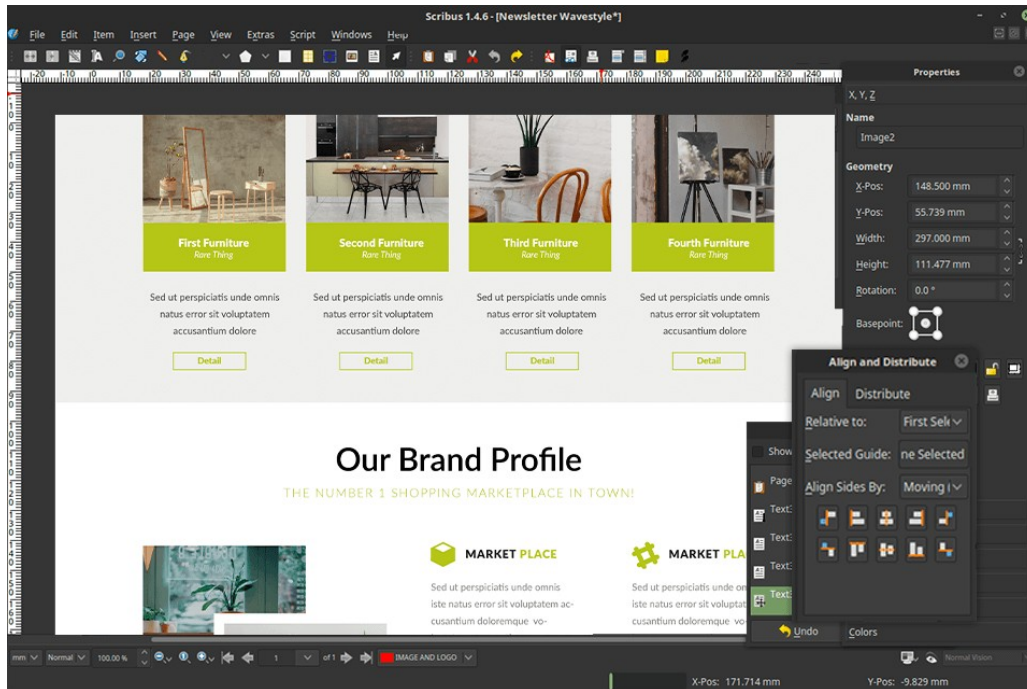
GIMP

Raster graphics editor



Scribus

Publishing (InDesign)



Blender

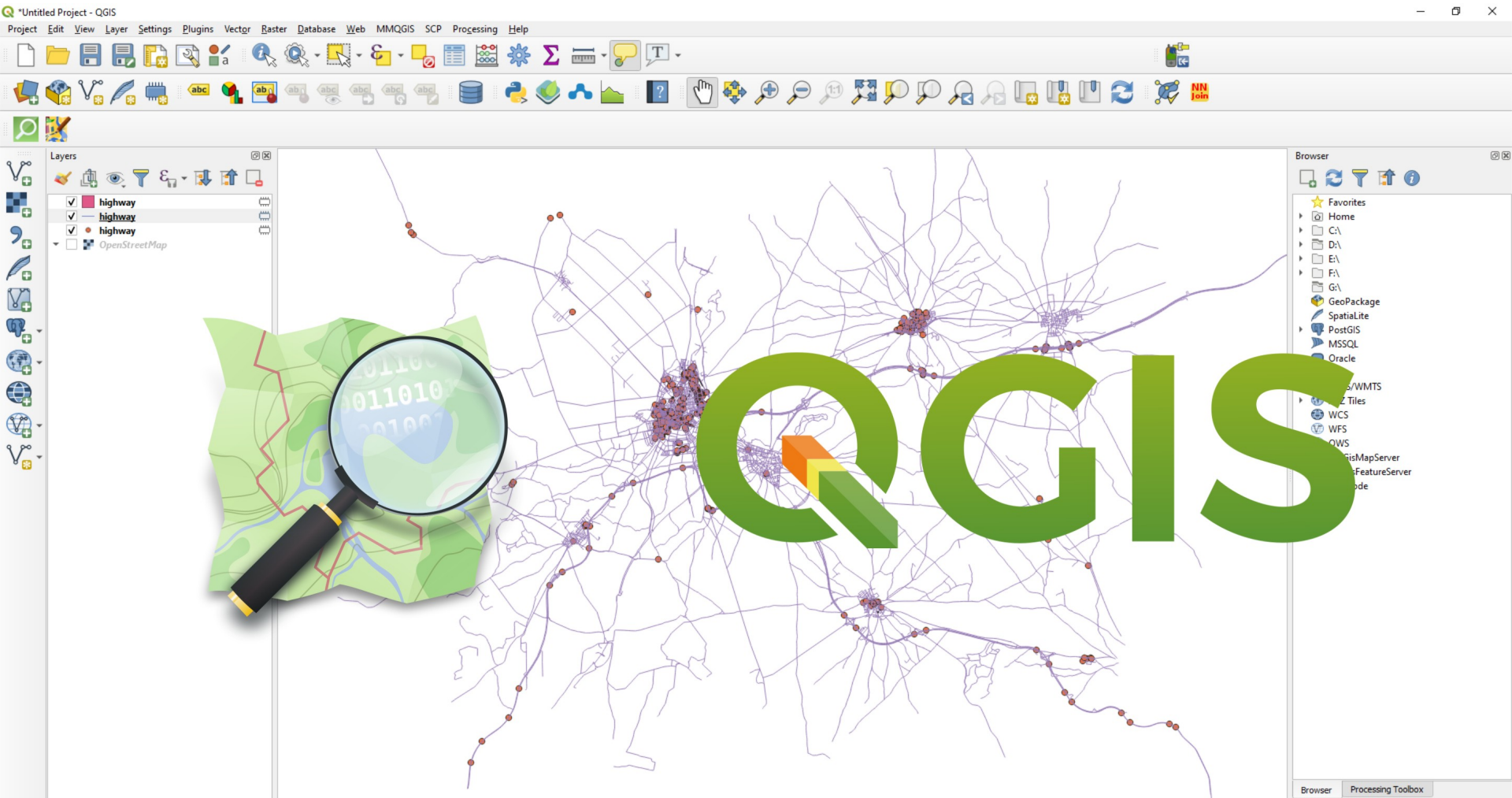
*3D Graphics, Modelling, Shading,
Animation, Rendering*

blender



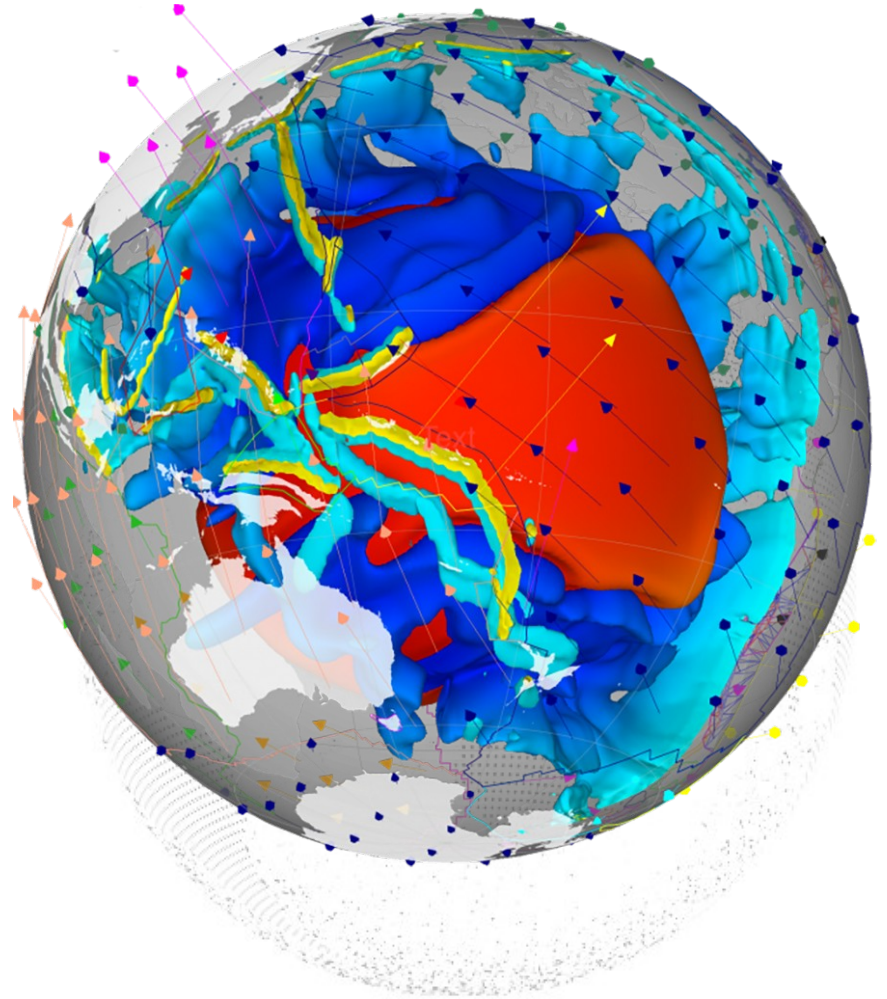
QGIS

Open source GIS



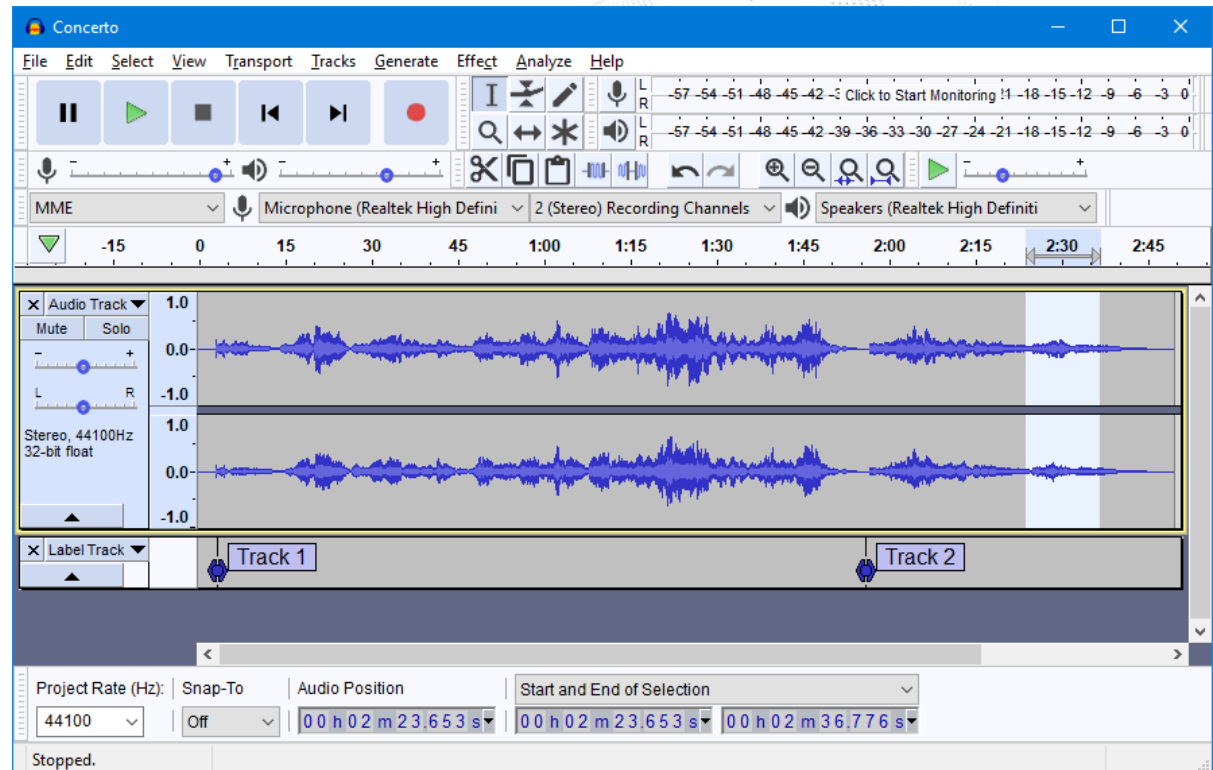
GPlates

Plate tectonic reconstructions



Audacity

Sound and music editor



Office

Open source



Hundreds of command line tools, e.g.

Multimedia:  **FFmpeg**

Images: 

Compiler: 

Document conversion:

